EMBEDDED / CONTROL / AUTONOMY

AEGIS AUTONOMY

PROJECT 02

Trustworthy autonomy under sensor uncertainty and failure.

A fault aware autonomy research platform investigating whether a vehicle can recognise deteriorating state trust and degrade safely before control decisions become unsafe.

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM / RESEARCH ARCHITECTURE

1 STAGE
Vehicle dynamics

→

2 STAGE
Estimator

→

3 STAGE
Fault monitor

→

4 STAGE
Degraded controller

THE PROBLEM TO SOLVE

Which monitoring and degraded control strategies reduce risk without triggering unusable false alarms when state estimates become unreliable?

TECHNICAL DISTINCTION

1. This is not another generic drone simulator; the differentiator is uncertainty aware supervision.
2. The value comes from controlled experiments, negative results and honest limits, not pretty trajectories.

WHAT THE EVIDENCE MUST SHOW

- ✓ Fault detection delay and false alarm rate
- ✓ Trajectory error and safety boundary violations
- ✓ Sensitivity to estimator miscalibration and correlated faults